

Computing Improper Integrals

1. Evaluate the following Type I improper integrals.

(a) $\int_1^{\infty} \frac{1}{\sqrt[4]{x}} dx$

(b) $\int_{-\infty}^{-2} \frac{1}{x^3} dx$

(c) $\int_4^{\infty} \frac{1}{x \ln x} dx$

(d) $\int_{-\infty}^{\infty} \frac{1}{x^2 + 1} dx$

2. Evaluate the following Type II improper integrals.

(a) $\int_0^1 \frac{1}{\sqrt[4]{x}} dx$

(b) $\int_{-2}^0 \frac{1}{x^3} dx$

$$(c) \int_0^1 \frac{\ln x}{x} dx$$

$$(d) \int_0^{1/2} \frac{1}{x \ln x} dx$$

$$(e) \int_2^4 \frac{1}{x(x-2)^2} dx$$

$$(f) \int_2^4 \frac{1}{x^2(x-2)} dx$$

3. Torricelli's trumpet (also called Gabriel's Horn) is the infinite 3D shape formed by rotating the region between $y = \frac{1}{x}$ and $y = 0$ for $x \geq 1$ around the x -axis.

Compute the volume of Toricelli's trumpet.

$$\begin{aligned} \text{Volume} &= \pi \int_1^\infty (r(x))^2 dx \\ &= \pi \int_1^\infty \left(\frac{1}{x}\right)^2 dx \end{aligned}$$

For surface area, use the comparison theorem.

$$\begin{aligned} \text{Surface Area} &= \int_1^\infty 2\pi r(x) ds \\ &= 2\pi \int_1^\infty \frac{1}{x} \sqrt{1 + \left(-\frac{1}{x^2}\right)^2} dx \end{aligned}$$

This was considered a major paradox during the 1600s. The strange nature of infinity and its properties caused much dispute and struggle among mathematicians and philosophers (including Galileo).